Skanda Koppula

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Work Experience

Google DeepMind, Staff Research Engineer *September 2019 - Present*

- Data and evaluations lead of video-to-3D capabilities research team. Responsible for synthetic data generation (Blender, Unreal, PyBullet), data quality monitoring and filtering, YouTube pseudolabeling strategies, efficient evaluation pipelines, and expanding to new inference hardware.
- Technical lead of vision research team: built Jax training setups for distributed model training, video-text pre-training, evaluations for video Q/A, and video+language data pipelines (2022-2024)
- Engineer on visual modeling research team: researched (1) methods for large-scale visual representation learning (2) multi-modal self-supervised learning objectives, and (3) architectures for dense visual tasks such as object detection, segmentation, and optical flow. (2019-2022)
- Selected publications:
  - Efficiently Reconstructing Dynamic Scenes One 🎯 D4RT at a Time (CVPR 2026, Best Paper Award Nominee, third author)
  - A Simple Recipe for Contrastively Pre-training Video-Language Encoders Beyond 16 Frames (CVPR 2024, first author)
  - Where Should I Spend My FLOPS? Efficiency Evaluations of Visual Pre-training Methods. (NeurIPS SSL Workshop 2022, first author)
  - TAPVid-3D: A Benchmark for Tracking Any Point in 3D. (NeurIPS 2024, first author)
  - Perception Test: A Diagnostic Benchmark for Multimodal Models. (NeurIPS 2023, 7th author)
  - For the full list, and open-source implementations, please see my website and GitHub!

NVIDIA Autonomy, PilotNet Team, Intern *May 2018 - September 2018*

- Added support for multi-camera vehicle feeds to the simulation and end-to-end driving model
- Decreased training time by 16% using custom data format to reduce disk reads and CRC overhead.

Google Acoustic Modeling Team, Intern *June 2017 - September 2017*

- Researched approach to visualize the memory contents and hidden state of recurrent neural networks used in character language models. Work published in ICASSP 2018.
- Tested kernel factorization as a way to improve cross-domain accuracy of neural language models.

Projects

ETH Zürich, Fulbright Research Fellowship, Researcher *September 2018 - June 2019*

- Developed CNN fine-tuning procedure to improve bit error resilience. Paired with approximate DRAM modules (voltage and timing scaled), this yielded a 29% power reduction and 5% speed-up while running CNN inference on CPU and GPU.

MIT Formula Student Driverless, Team Co-Founder *September 2017 - March 2019*

- Helped hire and manage new recruits for a team that built the entire autonomy stack of a Formula Student racecar. Closely directed technical efforts on the perception team, demonstrating an accurate LiDAR-stereo camera system in PyTorch for racetrack landmark localization.
- Won 2nd and 3rd place at the 2019 Formula Student Driverless Italy and Germany competitions.

Education

University College of London *Advisors: Prof. Gabriel Brostow, Andrew Zisserman*
Doctorate of Philosophy, Computer Science *Sept. 2023 - Present*

Massachusetts Institute of Technology
Masters of Engineering and BSc, Computer Systems, MEng GPA: 5.0/5.0 *Sept. 2013 - June 2018*